

Hardening Mixture-of-Agents: An Evidence-Bound Harness for Specialist Aggregation Pipelines

Sam Bobo

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Abstract

Mixture-of-Agents (MoA) pipelines route a task through several proposer models and synthesize their outputs through an aggregator, and report state-of-the-art results on preference-judged benchmarks. Every published gain, however, is measured by a preference judge, and nothing in that evaluation tests whether the aggregator verifies what it is given, whether proposer caution reaches the reader, or whether a corrupted proposer corrupts the answer. This paper presents an evidence-bound harness that hardens the MoA recipe against exactly those failures. The harness keeps the recipe’s proposer-aggregator structure and its measured preference lift, and adds five mechanisms: an empirical seating gate, engineered redundancy across proposers, re-consultation triggered by a mandated artifact rather than by any model’s self-assessment, an out-of-band channel that carries proposer qualification around the aggregator, and a charter of diagnostics that may never be optimized. Each mechanism is licensed by a pre-registered experiment on an auditable MoA instance (7 to 20B local models, more than fifty registered experiments) that falsified a premise the recipe depends on: the aggregator adopts a sole-sourced corrupted value at 0.90, an instruction to mark uncertainty installs only its named phrase, order-debiased judges penalize that phrase while rewarding aggregation, prose transports 0.11 to 0.33 of supplied qualification, and elicited self-assessment predicts behavior at chance. Under prospective single-factor re-tests, engineered redundancy halves corrupted-value adoption (0.900 to 0.450), the out-of-band channel carries 1.000 of proposer caveats against 0.174 through prose and changes reader decisions (+0.691 over a matched control), live re-consultation resolves contested quantities at 0.905 against 0.333, and the seating gate predicts in-pipeline defects (+0.632). Two of the harness’s own rationales were demoted by registered tests and one gate failed twice. Judged head-to-head against the unmodified pipeline under a two-family order-debiased protocol, the assembled harness shows no evaluable preference difference: it fixes what the recipe’s evaluation cannot measure at no cost on what it can.

1. Introduction

Multi-agent pipelines that aggregate several language models into one answer have become an accepted recipe. Mixture-of-Agents (MoA) [1] reports that a stack of open proposers plus an aggregator surpasses GPT-4o on AlpacaEval 2.0 (65.1% against 57.5% length-controlled win rate), and explains the result as collaborativeness: an aggregator shown several drafts produces a better response than any of them, even when some drafts are weak. Production variants cast the proposers as domain specialists and the aggregator as a lead writer that decomposes the problem, dispatches sub-questions, weighs the specialists’ contributions, and synthesizes a recommendation. In advisory settings, healthcare, legal, and financial analysis among them, the recipe is attractive precisely because specialists are expected to contribute knowledge and caution that a single generalist would lack.

The recipe’s evaluation does not test the story that motivates it. Every published MoA number is produced by an LLM preference judge shown two answers and

asked which is better. No published evaluation measures whether the aggregator checks what it is given, whether a corrupted proposer corrupts the final answer, whether the specialists’ caveats survive synthesis, or whether an instruction to the aggregator changes what it does rather than what it says. Established instruction-following and preference-optimization practice assumes that a system prompt asking the aggregator to be critical makes it critical, and that a preference judge rewards the result. Neither assumption had been tested at the pipeline level.

This paper presents an evidence-bound harness that hardens the MoA recipe against the failures its evaluation cannot see (Figure 1). The harness keeps the recipe’s structure, a set of proposer seats feeding a synthesizing lead, and keeps its one demonstrated gain, a preference lift from aggregation that replicates at auditable scale under an order-debiased protocol. It adds five mechanisms: an empirical seating gate that admits proposers by measured behavior, a planner that assigns every load-bearing quantity to at least two proposers, a re-consultation loop triggered by a mandated artifact

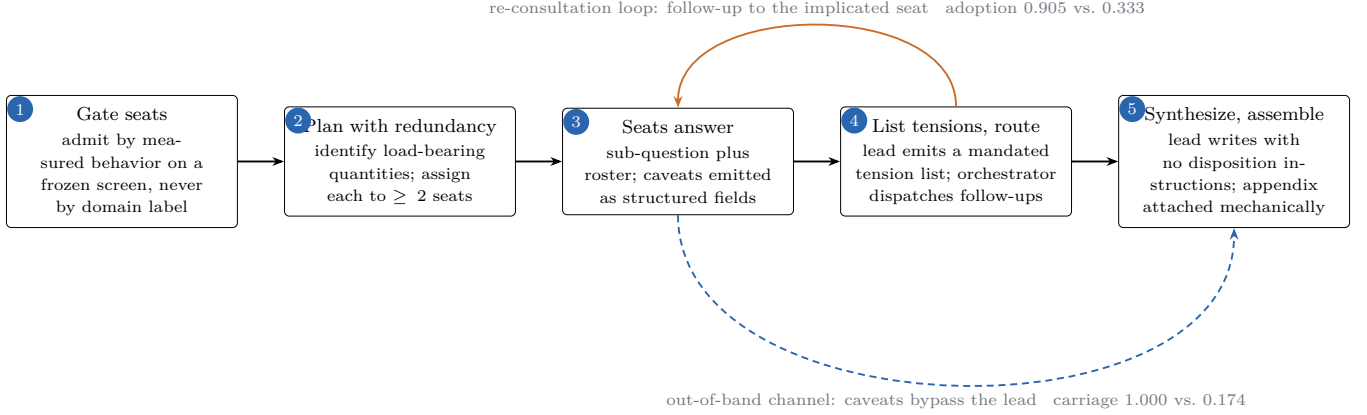


Figure 1: Overview of the harness. Steps 1 through 5 retain the proposer-aggregator structure of Mixture-of-Agents and add the five mechanisms of Section 4. The dashed blue lane carries proposer qualification to the final artifact without passing through the lead; the orange loop routes follow-up questions from the lead’s mandated tension list back to the implicated seat. Every edge in the full diagram (Figure 6) is labeled by its registered measurement.

rather than by any model’s judgment of its own competence, an out-of-band channel that carries proposer caveats, assumptions, and confidence to the final artifact without passing through the lead, and a charter of diagnostics that may never become optimization targets. No mechanism is included on rationale. Each is licensed by a pre-registered experiment, run on an auditable MoA instance of three specialist seats and one 20B aggregator across eighteen advisory scenarios, that falsified a premise the recipe depends on (Section 3).

The falsifications are specific. The aggregator selects among sources and never recomputes: a planted corrupted value reaches the answer 0/27 times when a clean co-source exists and is adopted at 0.900 when sole-sourced. An instruction to label estimates installs only its named phrase, in two thirds of outputs, with the behavior beyond the phrase unchanged under a synonym swap. Order-debiased preference judges reward aggregation (0.818 of decisive pairs against the same writer answering directly) and penalize the instructed phrase (0.261 and 0.317 against control), a two-link causal chain that places instruction tuning and preference optimization in direct tension at the surface where instructions act. Prose transports only 0.11 to 0.33 of the qualification supplied to it. Fine-tuned specialists carry per-model rather than per-domain signatures. And every elicited form of self-assessment, deferral, prior plausibility, and ownership, predicts behavior at chance.

Each harness mechanism was then submitted to a prospective single-factor test with its kill condition registered in advance (Section 6). Engineered redundancy halves corrupted-value adoption (0.900 to 0.450) and is causal. The out-of-band channel carries 1.000 of both-judge proposer caveats against 0.174 through prose, and

a delivered caveat flips a reader’s registered decision at 0.727 against 0.036 for a matched irrelevant control. Live re-consultation conveys the deciding fact in 21/21 dispatched replies and the lead adopts it at 0.905 against 0.333. The seating gate separates defecting from non-defecting seats with strict rank separation (+0.632). Two of the harness’s own rationales were demoted by informative nulls, one candidate routing rule was rejected by its confirmation experiment, and one delivery gate failed its frozen criteria twice and is recorded as not deployable. Judged head-to-head against the unmodified pipeline under a two-family order-debiased protocol, the assembled harness shows no evaluable preference difference (0.562 [0.375, 0.765]; 0.333 [0.083, 0.562]). That null is the paper’s central result: the harness fixes what the recipe’s evaluation cannot measure at no cost on what it can.

2. Problem Formulation

The aggregation recipe. A task q is dispatched to n proposer seats $P_1, \dots, P_n$, each returning a contribution $c_i = P_i(q_i)$ on a sub-question q_i . An aggregator A receives the concatenation and returns the final artifact $a = A(q, c_1, \dots, c_n)$. The recipe’s published metric is the share of pairwise comparisons in which a judge J prefers a to a baseline b . With the judge’s verdict $J(x, y) \in \{x, y\}$ and D the set of pairs on which the verdict is stable under reversal, the order-debiased preference share is

$$\text{pref}(a, b) = \frac{|\{(a, b) \in D : J(a, b) = J(b, a) = a\}|}{|D|}. \quad (1)$$

Section 5 shows that at auditable scale $|D|$ is a small minority of pairs, because raw first-position preference

is 0.85 to 0.88 regardless of judge.

Outcomes the recipe does not measure. Three quantities bear on whether the recipe delivers what it promises, and none appears in its evaluation. Adoption is the share of runs in which a value planted in some c_i appears in a ; the recipe assumes A verifies, so corrupted values should not be adopted. Transport is the relation between the count s of qualified sentences supplied across the c_i and the count y emitted in a , fitted within a corpus as

$$y = w s + c, \quad (2)$$

where the recipe assumes w near one. Trigger rate is the share of runs in which the pipeline recognizes that a contested quantity requires re-consultation; the recipe, where it re-consults at all, draws the trigger from a model’s report of its own confidence.

Compliance and behavior. An instruction to A is evaluated by an instrument that counts a construct in a . If the instruction dictates strings the instrument detects, the measurement M confounds compliance C with behavior B ,

$$M = C + B, \quad (3)$$

and a perfect detector still counts an echoed phrase. Separating C from B requires a design in which the dictated string varies while everything else is held fixed; Section 3 supplies one.

What a hardening must show. A harness that modifies the recipe should reduce corrupted adoption, raise transport of qualification to the reader, and draw its trigger from something that fires, while leaving Equation 1 no worse. Each claim must survive a registered test with the failure condition stated first.

3. Failure Modes of the Aggregation Recipe

The recipe rests on premises its evaluation never tests. Each was tested on the auditable instance of Section 5. This section reports the measurements; Table 1 maps each failure to the harness mechanism it licenses.

3.1. Instructions Install Only Their Named Phrase

The recipe’s aggregate-and-synthesize prompt asks the aggregator to be critical, preserve accuracy, and weigh its sources. Whether such instructions change behavior or only wording was tested by Equation 3’s required design. Three arms were run on 18 cases with 7 repeats (378 runs): a minimal writer prompt as control, and two treated arms appending a byte-identical clause differing in exactly three words, instructing the writer to label estimates with the phrase “modeled at” (form X) or “taken to be” (form Y). Form X had been dictated through months of the program’s prompts; form Y was chosen the day before the run and appears nowhere in 2M characters of archive. Each instructed arm produced its own

phrase in about two thirds of outputs (0.667, 0.698); control produced neither; cross-form leakage was 0/252 (Figure 2, left). With each arm’s own phrase subtracted, the validated-judge construct rate was unchanged: form X moved from 0.341 to 0.365 ($[-0.079, +0.111]$), form Y from 0.341 to 0.317 ($[-0.103, +0.087]$), with emission flat across arms (Figure 2, right). The registered minimum detectable effect was +0.224, so the result licenses “no form-independent effect of at least 0.22” rather than zero. A months-old house phrase and a day-old synonym performed identically: whatever the instruction engages, it is not a learned disposition.

3.2. Preference Rewards Aggregation and Penalizes the Phrase

Three pairwise comparisons were run on the same 18 cases (126 pairs each; both orderings; primary judge gpt-oss:20b, which also wrote both sides of every pair, making self-preference symmetric). The aggregated pipeline beat the same writer answering directly in 0.818 [0.718, 0.919] of 44 decisive pairs. Arm lengths differed by 1.4%, the share among near-length-matched pairs was 0.815, and the winner had more domain frameworks per kilocharacter in only 0.409 of decisive pairs, so the lift is neither length nor framework density; its composition is unidentified by the available feature set. The phrase-instructed arms, which differ from control only in the phrase, lost to control at 0.261 [0.140, 0.388] (form X) and 0.317 [0.152, 0.480] (form Y), against a registered prediction of the opposite (Figure 3). A second judge from a disjoint family (a 30B Qwen3 MoE, selected blind to direction) replicated the lift (0.788 [0.645, 0.917]) and the form-Y penalty (0.217 [0.125, 0.333]); the form-X penalty agreed in point estimate (0.276) with an interval spanning 0.5 at $n=29$ and is reported as not replicated at that power. Chained with Section 3.1, both links causal: an epistemic instruction installs only a surface form, and a preference judge penalizes that form. A pipeline tuned toward judge preference will shed exactly the marking that instructions install while the underlying disposition, untouched by the instruction, stays where it was.

3.3. The Aggregator Selects Sources and Never Verifies

With arithmetic claims planted in proposer texts, a corrupted value reached the answer 0/27 times when one proposer was corrupted and a clean one remained, 5/27 when every proposer was corrupted, and 12/27 when the working was also stripped (Figure 4). A later single-factor experiment sharpened the bare case: a sole-sourced corrupted value is adopted at 0.900. The aggregator prefers uncorrupted sources when they exist and has no independent hold on the content when they do not. MoA’s observation that the aggregator’s output overlaps most with the best proposals [1] is the same phenomenon from the benign side; the planted-error view

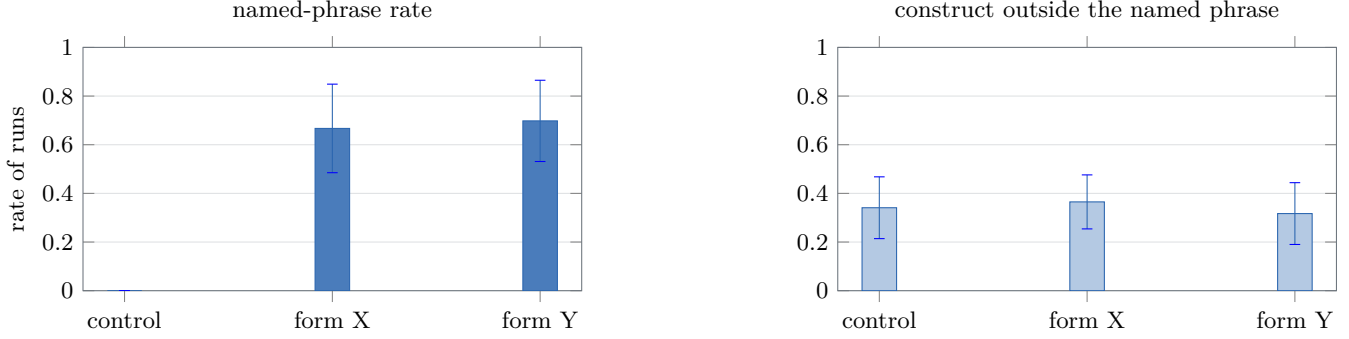


Figure 2: The phrase-swap experiment (378 runs, cluster-bootstrap 95% CIs over 18 cases). Left: each instructed arm produces its own named phrase in about two thirds of runs; control never produces either, and cross-form leakage is 0/252. Right: the validated-judge construct rate outside the named phrase is statistically indistinguishable across all three arms. The instruction’s entire measurable effect is the phrase it names.

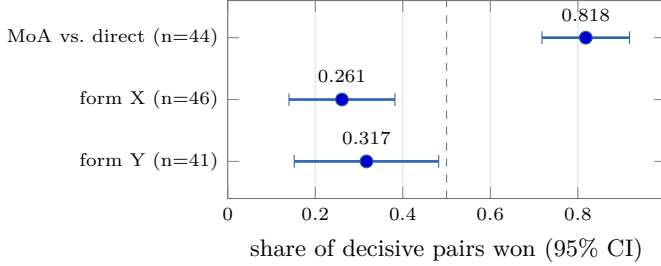


Figure 3: The preference experiment under Equation 1. The aggregated pipeline beats the matched-length direct baseline; both phrase-instructed arms lose to their own control, in counterbalanced forms whose only difference from control is the phrase. The dashed line is indifference.

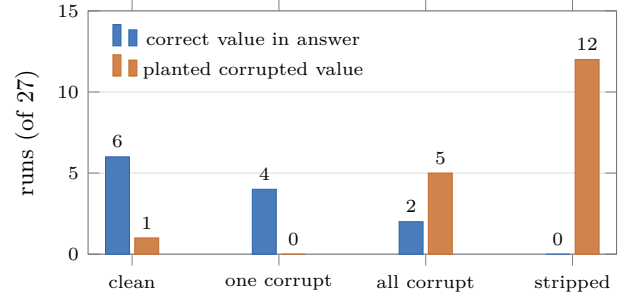


Figure 4: The source-selector gradient (planted arithmetic errors; exact match on known values). As clean sources are removed, adoption of the planted wrong value rises monotonically through 1, 0, 5, and 12 runs while correct values vanish.

adds the failure boundary, and the claim of improvement even from worse inputs fails at the fact level.

3.4. Corroboration Is Ignored and Transport Is Weak

Claims advanced by two or more independent proposers transport to the final answer no more often than single-proposer claims (0.487 against 0.524; difference CI $[-0.153, +0.080]$). Epistemic qualification fares worse. Across 144 runs, two writers, and two corpora, the within-corpus slope w of Equation 2 runs from 0.105 to 0.326, three of four fits excluding zero, with the extension-corpus fit replicating on nine cases the original corpus never contained (Figure 5). The writer emits roughly one sixth to one third of the caution it is given, the fraction is a property of the writing step rather than of one model, and the writer also invents qualification at 0.333 $[0.138, 0.609]$ on inputs confirmed to contain none.

3.5. Specialists Carry Per-Model Signatures

Two medical fine-tunes of the same base model do not share a domain signature: one shifts in-domain framework density $+0.232$ $[+0.072, +0.412]$ over the base, the other $+0.109$ $[-0.056, +0.300]$, and a second lin-

age shows -0.041 $[-0.099, +0.025]$. A descriptive check found two same-domain seats no more distant from each other than one seat is from itself across samples, consistent with the Self-MoA result [2] at benchmark scale. Seating a model for its domain label has no measured warrant.

3.6. Self-Assessment Predicts Behavior at Chance

Seats given a roster and a structured deferral token route to the correct colleague when they defer (10/11, 0.909 $[0.623, 0.984]$) but defer about as often in-domain as out-of-domain (0.083 against 0.306; difference CI $[-0.056, +0.528]$). A model’s own plausibility rating predicted which planted values it would adopt at 0.544. A stable, near-unanimous elicited map of which seat owns each quantity predicted arbitration on fresh items at exactly 0.500 against a descriptive 0.895. In every consultation design measured, the mechanism worked when fired and the firing was rare: the lead used a routed clarification in 7/7 runs but named the tension warranting it at 0.34, and across three experiments the trigger rate ranged from 0.208 to 0.583 by batch. Self-

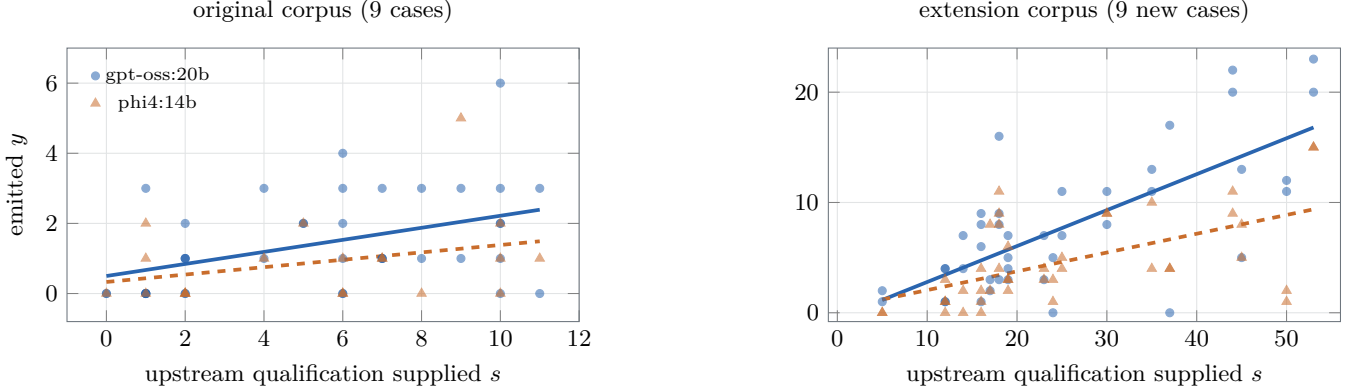


Figure 5: The transport coefficient of Equation 2, 144 runs over 18 cases and two writers; each point is one run, both quantities counted by the validated two-judge sentence protocol, lines are within-panel least-squares fits. Within-panel slopes: gpt-oss 0.172 [+0.052, +0.297] and 0.326 [+0.207, +0.434]; phi4 0.105 [−0.031, +0.273] and 0.170 [+0.067, +0.265]. Panels are separated because the corpora occupy different supply regimes (mean s of 5.4 against 25.3).

assessment, the faculty that escalation designs rely on, was the one faculty the program never found.

3.7. Accuracy Is Unstable at This Scale

On a 60-item self-verifying battery the aggregated pipeline and the bare writer were indistinguishable at a 0.92 to 0.97 ceiling. Two registered follow-up batteries (36 multi-step computation items and 30 rule-interaction items with script-computed answers, under baseline-only difficulty screens) placed the direct writer at 0.977 and 0.983. Under the pre-registered two-strikes rule the question is resolved as unstable for lack of headroom. The harness claims no accuracy gain and is not designed to produce one.

4. Evidence-Bound Harness

The harness retains the recipe’s structure and preference lift and adds the five mechanisms of Table 1. Figure 6 shows the assembled pipeline with every edge labeled by its registered measurement.

4.1. Empirical Seating Gate

Because signatures are per-model (Section 3.5), no seat is chaired for its category label. A candidate passes a frozen gate consisting of a degeneracy screen (at least 800 visible characters on screen items; one archive fine-tune produced 12/12 degenerate probe runs), a tuned-minus-own-base signature probe, and a format smoke test at production token budgets. A generalist under a role prompt is a legitimate seat if it passes. Verdicts are re-computed at seating time, never imported from the archive (Section 6.1).

4.2. Isolation on Cost Grounds

Seats receive a self-contained sub-question and a one-line roster; sibling outputs are not forwarded. The rule was adopted on a contamination rationale that a registered

Failure of the recipe	Measurement	Mechanism licensed
Instructions only (§3.1)	install their phrase unchanged, MDE 0.22	no disposition instructions to the lead
Judges penalize the phrase (§3.2)	0.261, 0.317 vs. control	out-of-band channel; charter
Aggregator selects, never verifies (§3.3)	0.900 sole-sourced; 0/27 with co-source	engineered redundancy
Corroboration ignored; transport weak (§3.4)	$w = 0.11$ to 0.33	out-of-band channel
Per-model signatures (§3.5)	+0.232, +0.109, −0.041	empirical seating gate
Self-assessment at chance (§3.6)	0.544; 0.500	artifact-triggered re-consultation
Accuracy at ceiling (§3.7)	0.92 to 0.98 direct	no accuracy claim

Table 1: Each failure mode of the recipe, its measurement, and the harness mechanism it licenses.

test later withdrew (Section 6.6); it stands because sibling context costs approximately 10k input characters and 14% longer outputs per seat while purchasing no measured change in lane discipline.

4.3. Engineered Redundancy

Because the lead selects among sources and never verifies (Section 3.3), protection is coverage. The planner identifies the load-bearing quantities in the decomposition and writes each into the sub-questions of at least two seats. Expected protection is the product of identi-

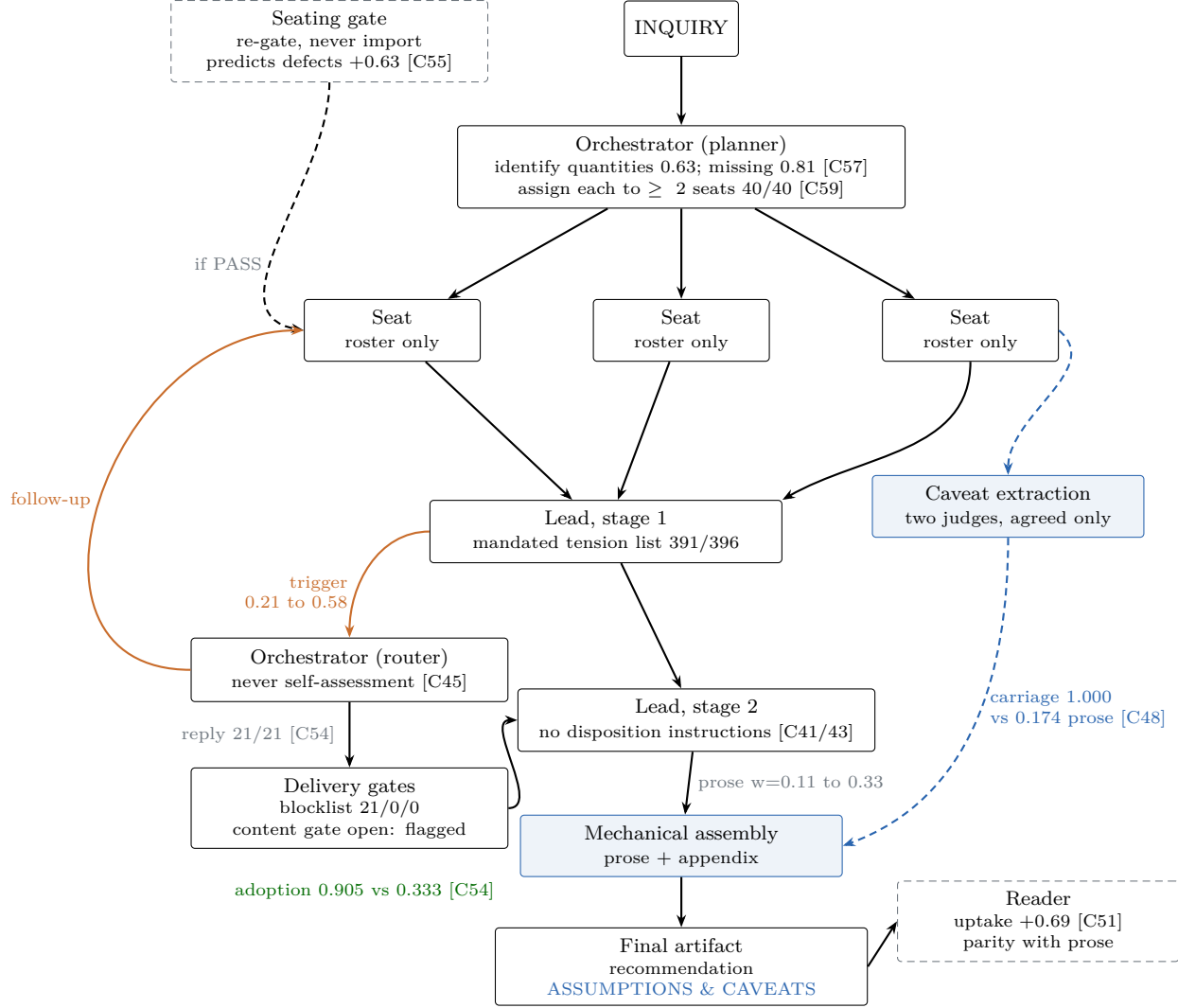


Figure 6: The harness on one inquiry, with every edge labeled by its registered measurement (bracketed codes are ledger entries). Dashed blue: the out-of-band channel for epistemic qualification, which the writer never sees. Orange: the orchestrator-routed re-consultation loop, bounded by its measured trigger rate. Chairs are earned by the gate, never by category label, and engineered overlap between seats is where error immunity resides.

fication coverage and the selection effect,

$$\begin{aligned} \Pr[\text{clean adopted}] &\approx \Pr[\text{identified}] \\ &\times \Pr[\text{clean selected} \mid \text{covered}], \end{aligned} \quad (4)$$

and is stated as expected value only (Section 6.2).

4.4. Artifact-Triggered Re-consultation

Because self-assessment predicts behavior at chance (Section 3.6), no model decides whether it needs help. The lead runs in two stages: a mandated tension list first (391/396 archived runs produce one when required), then synthesis. The orchestrator parses the list and dispatches a follow-up sub-question to the implicated seat. Replies pass through a fabrication blocklist seeded from committed classifications and a content flag for

replies adding nothing beyond the seat’s first contribution; flagged replies are delivered with the flag rather than dropped, because the content gate did not meet its frozen criteria (Section 6.6).

4.5. Out-of-Band Epistemic Channel

Because prose transports a small fraction of supplied qualification (Section 3.4) and preference judges penalize the visible marking that instructions install (Section 3.2), caveats, assumptions, and confidence never route through the writer. Seats emit them as structured fields; a two-judge extraction retains only sentences both judges agree carry qualification; and the harness assembles them mechanically into an appendix of the final artifact. The writer receives no disposition instructions at

all.

4.6. Goodhart Charter

Trigger rates, hedge counts, compliance shares, transport slopes, and preference scores on epistemic content are diagnostics and may never be optimization targets; the optimizable signal must live outside the pipeline’s own text. Each forbidden entry cites the experiment that showed what optimizing it would manufacture (Appendix A).

5. Experimental Setup

Pipeline and cases. The auditable instance is a two-layer MoA: three proposer seats (prompted or domain-fine-tuned 7 to 8B models, or role-prompted samples of the writer) whose contributions are concatenated for a 20B mixture-of-experts writer (gpt-oss:20b). A lead planner performs step-back decomposition and dispatches a self-contained sub-question to each seat. The case battery is eighteen strategic advisory scenarios spanning health-care, legal, and finance content; individual experiments use subsets of 6 to 18 cases, and every confidence interval is a cluster bootstrap over cases.

Registration. Every experiment is registered before it runs with its hypotheses, falsification conditions, attainability computation (the fraction of runs in which the outcome variable can fire, computed from existing data), and consequences. Verdicts enter a ledger distinguishing live, provisional, withdrawn, and blocked claims; three program-level audits re-classified earlier results, and nulls are cited only at the strength their audit class permits. Registrations are committed before first runs, so temporal ordering is verifiable from version history.

Instruments. Construct presence is scored by a two-judge, batched, sentence-level protocol whose definitions name no phrase; agreement on the deployed corpus is 0.910 (14,088/15,485 decisions). A frozen dictation registry of every phrase any prompt places in a model’s mouth (125 entries) gates all prompts, with a measured over-attribution of zero across 2,907 clean-provenance spans. Preference is judged in both orders with agreement required (Equation 1); raw first-position preference is 0.85 to 0.88 across four judge families, and debiasing converts 60 to 95% of pairs to ties (Appendix C).

Baselines. The unmodified pipeline (the recipe as published, with the MoA aggregate-and-synthesize prompt or a minimal lead prompt), the direct writer (same model, same cases, prompt minus the sentence referencing contributions), and, for re-consultation, a scripted-reply ceiling and a length-matched contentless-reply control.

Metrics. Corrupted and clean adoption (exact match on planted values); carriage (share of both-judge caveat

sentences reaching the final artifact); uptake (share of a reader model’s registered two-token decisions flipped by a delivered caveat, with direction-balanced items); defect rate (share of a seat’s contributions violating its lane); trigger rate; and the order-debiased preference share of Equation 1.

6. Results

Each mechanism received its own registered single-factor experiment with kill conditions stated in advance. Table 2 collects the main results against the unmodified pipeline and Figure 7 plots them; the subsections give the measurements.

6.1. Seating Gate

Ten local candidates were gated on the frozen degeneracy screen using two screen items disjoint from the pipeline case set, with verdicts committed before any pipeline run, and every candidate was then seated on the pipeline cases. Both gate-fail candidates defected in the pipeline (0.333 and 0.944 of contributions) while all eight gate-pass candidates ran at 0.000 to 0.056; the pooled contrast is +0.632 [+0.549, +0.722]. Gate verdicts age: one model that passed the archive-era screen failed the fresh re-gate and then defected in-pipeline at 0.333. Format compliance is a separate axis: the legal fine-tune failed the format smoke test 0/4 yet seated defect-free.

6.2. Planner Chain

The planner was the one mechanism never exercised before the re-test campaign; every prior redundancy result used human-planted assignments. The orchestrator recalls load-bearing quantities at 0.633 [0.506, 0.758] against analyst-authored labels (a thirty-fold specificity separation over a mismatched-case control), unchanged by seeing the specialists’ contributions, and best on missing quantities (0.813 against 0.531 for stated figures). Given the identified quantity, the planner writes it into at least two sub-questions in 40/40 plans, live seats convey briefed values at 0.85 to 0.95, and the writer uses the clean co-source end-to-end (+0.425 [+0.325, +0.525] against the planted +0.500). A halted first attempt yielded a channel constraint: privately briefed facts convey only into task-relevant assignments (2/8 orthogonal against 21/21 pointed), which the corrected planner input satisfies by construction. Equation 4 therefore evaluates to approximately 0.63 times the selection effect.

6.3. Engineered Redundancy

Assigning the contested quantity to a second seat, as the sole factor, halved corrupted-value adoption from 0.900 to 0.450 ([+0.200, +0.725]) and raised clean-value adoption from 0.000 to 0.500 ([+0.175, +0.825]). This is selection rather than dilution. Protection is bimodal across items; two candidate boundary mechanisms were

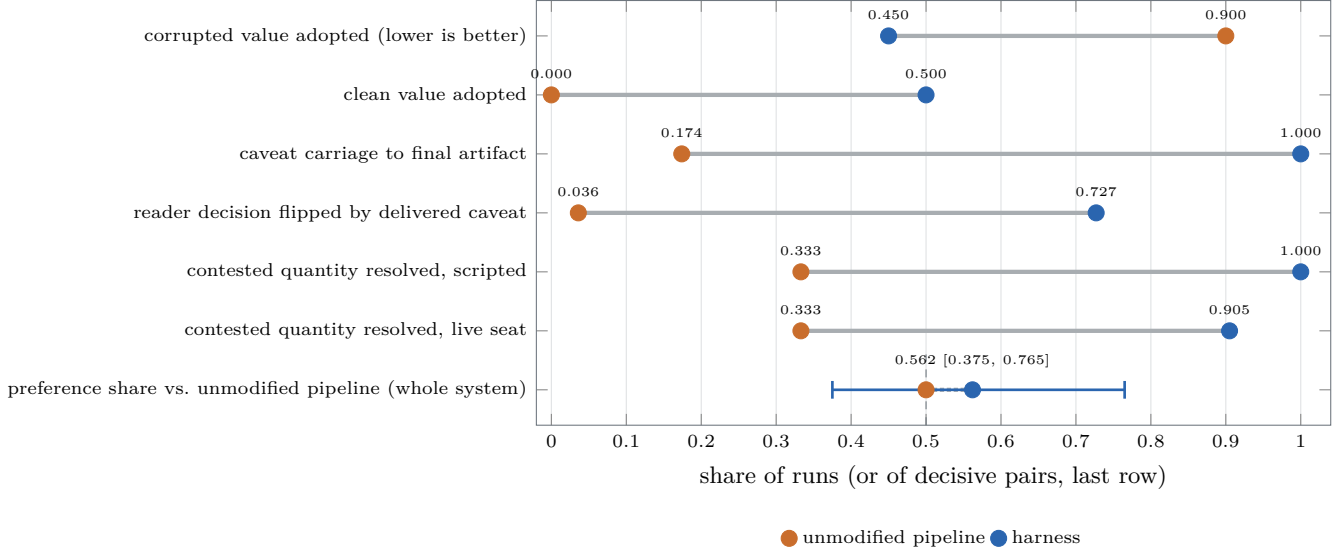


Figure 7: Outcomes before and after the harness. Each of the first six rows is one registered single-factor experiment in which the unmodified pipeline (orange) and the harness mechanism (blue) were measured on the same outcome; the rows are separate experiments, not one before-and-after run of the whole system. The last row is the only whole-system comparison, the order-debiased head-to-head of Section 7, with its 95% interval spanning the dashed indifference line. Corrupted-value adoption is the one row where lower is better. Values and intervals are those of Table 2.

Outcome	Mechanism under test	Unmodified pipeline	Harness	Verdict
Corrupted value adopted (sole-sourced) ↓	engineered redundancy	0.900	0.450 [+0.200, +0.725]	survived (causal)
Clean value adopted ↑	engineered redundancy	0.000	0.500 [+0.175, +0.825]	survived
Clean co-source used end-to-end ↑	planner, live	planted +0.500	+0.425 [+0.325, +0.525]	survived
Caveat carriage to artifact ↑	out-of-band channel	0.174	1.000	survived
Reader decision flipped by caveat ↑	out-of-band channel	0.036 (irrelevant ctrl.)	0.727	survived
Contested quantity resolved (scripted) ↑	re-consultation	0.333	1.000 (7/7)	survived
Contested quantity resolved (live seat) ↑	re-consultation	0.333	0.905	survived, one halt
Seat defect rate, gate-pass vs. gate-fail ↓	seating gate	0.333, 0.944 (fail)	0.000 to 0.056 (pass)	survived
Preference share vs. unmodified pipeline	assembled harness	0.500 (reference)	0.562 [0.375, 0.765]	no evaluable difference

Table 2: Main results. Each row is one registered single-factor experiment; shaded cells mark the harness where its interval excludes the baseline. The final row is the head-to-head of Section 7, in which no difference is evaluable and no cell is shaded.

registered and rejected (Section 6.6), and the boundary is recorded as open.

6.4. Out-of-Band Channel

The appendix delivers 1.000 of both-judge caveat sentences against the 0.174 that survives prose transport, with the preference cost not evaluable at power (0.510 [0.267, 0.754], 65/116 ties). A decision-relevant caveat delivered appendix-only flips a reader model’s registered two-token decision in 0.727 of reads against 0.036 for the identical appendix block carrying an irrelevant caveat, with direction-balanced items so that generic caution cannot masquerade as uptake, and at parity with weaving the same sentence into prose (Figure 8). The channel that preference judging cannot see changes decisions.

6.5. Routed Re-consultation

Scripted replies. Three arms over six planted tensions whose deciding fact was withheld from round one: no

clarification; a length-matched clarification responsive in style but containing no deciding fact; and the same clarification containing it. On runs where the tension was named, the lead adopted the fact-implied resolution in 7/7 informed runs against 0.333 in control (difference CI [+0.200, +1.000]) and 0.200 under the contentless clarification ([+0.545, +0.947]). The information does the work rather than the ritual, and the mechanism operates through the one capability the lead demonstrably has, preference among overlapping sources, by manufacturing an uncorrupted source at the contested quantity.

Live seat. Two further experiments replaced the scripted reply with a real seat. The first buried the deciding fact in the seat’s own round-one contribution and halted at its registered pilot gate, with the tension named in only 0.208 of runs. The second restored the original round-one bytes and supplied the fact only as the seat’s private working notes at reply time. The live seat

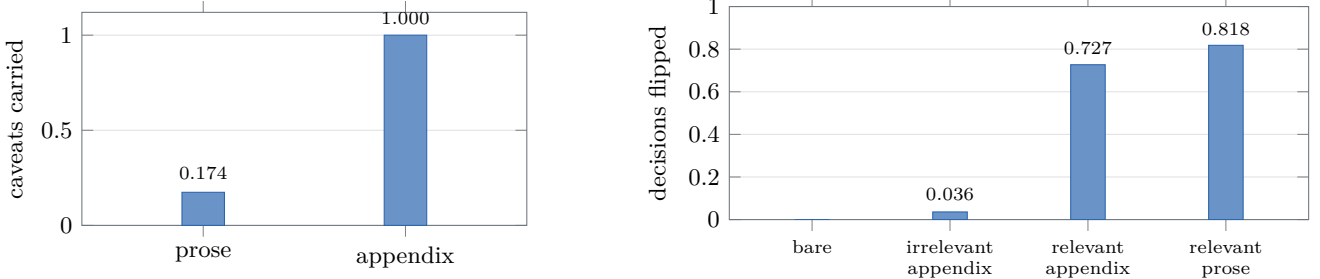


Figure 8: The out-of-band channel, both halves. Left: share of both-judge caveat sentences reaching the final artifact through writer prose against the mechanically assembled appendix (byte-identical prose in both arms). Right: share of registered two-token reader decisions flipped by a planted decision-relevant caveat, by delivery arm (11 items, 440 reads). Uptake is $+0.691$ [$+0.455$, $+0.891$] over the irrelevant-appendix control and at parity with in-prose placement ($+0.091$ [-0.036 , $+0.218$]).

	When triggered	Trigger rate
lead uses a clarification	7/7	0.34
seat routes a deferral	0.909 correct	0.306
briefed seat conveys, live	21/21 conveyed	0.208 to 0.583

Table 3: Re-consultation is recognition-limited rather than capability-limited, which is why the trigger is drawn from a mandated artifact and never from self-assessment.

conveyed the fact in 21/21 dispatched replies, and the lead adopted the fact-implied resolution at 0.905 against the archived control’s 0.333 ($+0.571$ [$+0.139$, $+0.912$]), within -0.095 [-0.227 , 0.000] of the scripted ceiling. In 4/21 live replies the seat embellished the correctly conveyed fact with fabricated legal authority and the lead adopted regardless; a registered follow-up bounded the rate at 0.100 [0.047 , 0.201] with the fact in hand and 0.150 [0.081 , 0.261] without it, concentrated on one trigger topic (15/20 there, 0/100 elsewhere) and name-stable across samples, which licenses the blacklist filter.

The trigger. Table 3 collects the three consultation designs: each works when fired and fires rarely on planted tensions. On organic inquiries the picture differs: in the head-to-head experiment’s 54 live pipelines, the orchestrator found a role-named tension and dispatched on 0.85 of runs.

6.6. Ablations and Rejected Components

Isolation. Making byte-fixed sibling contributions visible under production prompts moved off-domain framework density by -0.009 [-0.048 , $+0.028$] against a pre-registered minimum detectable difference of $+0.05$ to $+0.08$, with in-domain density unchanged. The contamination rationale is withdrawn and the rule stands on cost. A registered secondary measure pointed the other way (-0.033 [-0.053 , -0.013] off-domain terms when siblings were visible) and is recorded as a candidate mechanism, not claimed.

Ownership routing map. A stable, near-unanimous

elicited map of which seat owns each quantity, exactly the pattern a framework would ship as a routing table, predicted arbitration at 0.500 on fresh items against 0.895 descriptive, with a minimum detectable difference of 0.68. It is not in the harness.

Prior plausibility. A model’s own plausibility rating predicted which planted values it would adopt at 0.544 and was rejected as a boundary mechanism for redundancy.

Content gate. The rule that follow-up replies adding nothing should be dropped failed two registered attempts on complementary halves: a two-judge semantic gate passed vacuous fillers (3/6 dropped), and an extract-then-verify gate that fixed vacuity (6/6) rejected 52% of genuine live replies. Under the frozen commitment there is no third attempt without purpose-built labeled data; the deployed posture is flagged delivery. The fabrication blacklist, by contrast, validated at 21 true hits, 0 false positives, and 0 misses over the 141-reply archive and is deployed.

6.7. Generalization

Three results extend beyond the conditions that produced them. The preference lift and the form-Y penalty replicate under a judge from a disjoint model family (Section 3.2). The transport slope replicates on an extension corpus of nine cases the original corpus never contained, for both writers (Figure 5). And the trigger scarcity measured on planted tensions (0.208 to 0.583) gives way to 0.85 dispatch on organic inquiries, so the harness’s coverage bound on natural problems is set by the identification rate of Equation 4 rather than by the trigger.

7. Head-to-Head Preference Evaluation

With every mechanism separately verdicted, the remaining question was whether the assembled harness costs anything on the recipe’s own metric. The harness’s final artifact was judged against the unmodified pipeline’s under Equation 1 with two judge families (108 pairs). The

result is no evaluable difference: primary-judge share 0.562 [0.375, 0.765], replication 0.333 [0.083, 0.562], both spanning 0.5, with large effects excluded in both directions. The harness’s preference standing is inherited from aggregation, which both artifacts share; its mechanisms were never preference plays, and shipping them (five caveats per artifact, a follow-up round, restructured seats) costs nothing detectable. The appendix’s within-pair attribution is mildly positive (+0.073 toward the harness).

The null is the correct outcome rather than a short-fall. Preference judging at this scale is majority reading-order, penalizes visible caution, and cannot see whether a caveat reached the reader or whether a corrupted value was adopted. The failures the harness fixes are, by construction, invisible to the metric the recipe reports.

8. Related Work

Aggregation pipelines. MoA [1] is the recipe this paper hardens. Li et al. [2] showed that aggregating repeated samples of the single best model outperforms mixed MoA on AlpacaEval, converging from the benchmark side with the per-model signature result of Section 3.5. Wang et al. [6] report that a single strongly prompted agent can match multi-agent discussion, consistent with the finding that the one retained intervention routes information rather than deliberation.

Judge pathologies. Position and verbosity bias in LLM judges [3], length correlations under preference optimization [4], and sycophancy under preference pressure [5] motivate the order-debiased two-family protocol and sharpen into the measured phrase penalty of Section 3.2.

Instruction following and pre-registration. The phrase-swap design is, to the author’s knowledge, the first to test an epistemic instruction by swapping its named surface form under otherwise byte-identical prompts. No prior work is known that pre-registers each component of a multi-agent architecture and reports the failures alongside the survivals. The complete registration ledger, every deviation and verdict, and the measurement kit are released with the project repository (Appendix B).

9. Conclusion

The Mixture-of-Agents recipe delivers a preference lift and rests on a mechanism story its evaluation never tests. Tested under pre-registration at auditable scale, the story fails: the aggregator selects rather than verifies, instructions install phrases that judges penalize, prose loses most of the qualification it is given, and self-assessment predicts behavior at chance. The harness presented here accepts the recipe’s structure and lift, adds five mechanisms each licensed by one of those failures, and submits every mechanism to a prospective test

with the kill condition written first. The mechanisms that survived did so with causal or predictive effect sizes; two rationales were demoted, one candidate rule was rejected, one gate is recorded as not deployable, and the assembled harness is indistinguishable from the recipe on the recipe’s own metric. That is the hardening the paper proposes: not a better score, but a pipeline whose every component can say which experiment it survived.

Impact Statement

This work concerns the reliability of multi-agent language model pipelines in advisory settings, where a corrupted input adopted without verification or a caveat lost in synthesis can reach a decision maker. The harness is intended to reduce both failure modes and to make their residual rates measurable in deployment. The findings on preference judging carry a caution for evaluation practice: single-ordering LLM-judge comparisons at the studied scale predominantly measure reading order, and preference-trained pipelines can be expected to shed visible epistemic marking. All experiments used open models on local hardware with synthetic advisory scenarios; no human subjects or personal data were involved. Results are bounded to 7 to 20B models, advisory-domain English, and small case counts, and should be retested at larger scale before being relied on in deployment.

References

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A. Supplementary Discussion

The Goodhart charter. The following quantities are diagnostics and may not be optimization targets. Trigger rate: raising it blindly manufactures the contentless-consultation condition of Section 6.5, in which a length-

matched reply with no deciding fact moved adoption to 0.200. Hedge counts and compliance shares: raising them produces the dictated phrase of Section 3.1, which Section 3.2 shows judges penalize. Transport slope: optimizing it against the writer’s own text rewards invented qualification, which the writer already produces at 0.333 on inputs containing none. Preference score on epistemic content: the instrument is majority reading-order and penalizes visible caution. The optimizable signal must live outside the pipeline’s own text, in verified outcomes where ground truth exists.

Operating economics. Everything downstream of the trigger is measured and works. On planted tensions the trigger fires at 0.208 to 0.583 by batch; on organic inquiries dispatch fired on 0.85 of pipelines. Sibling isolation saves approximately 10k input characters and 14% output length per seat. The harness’s telemetry doubles as a replication battery: every deployment yields transport estimates, trigger telemetry, phrase-swap slots for proposed instructions, and preference-versus-verified divergence wherever ground truth exists, each with its attainability gate.

Open risks. Per-item protection is bimodal with the boundary mechanism unknown after two rejected candidates. The fabrication bound covers adversarial case citations on one trigger topic only, and the seat never admits the gap (0/20 sampled), so the bound is a floor rather than a ceiling on confabulation. The preference cost of the appendix is bounded but not evaluable at current power. The pipeline used no tool calling or retrieval, and the paper makes no claim about how those would interact with the harness.

B. Experimental Details

Phrase-swap. 18 cases, 7 repeats, 378 runs. Treated clause: “When you state a number that is an estimate rather than an established fact, label it using the phrase ‘modeled at’ ” (form X) or “...‘taken to be’ ” (form Y). Form X: 11 registry entries across 9 prompt symbols. Form Y: zero archive occurrences; spontaneous rate 0/249 clean-provenance runs for both forms. Emission 43.7 to 46.6 sentences per run across arms. A harness gate refused to run unless each arm contained exactly its own form and no other registry phrase.

Preference. 126 pairs per comparison, both orderings, decisive only when the same side wins both. Primary judge gpt-oss:20b; replication judge a 30B Qwen3 MoE selected from three candidates on parse and decisive rates only. Decisive counts: 44, 46, 41 (primary); 33, 29, 23 (replication). Arm mean lengths 7,757 against 7,649 characters for the lift comparison.

Source selection. 27 runs per arm; planted arithmetic errors; exact match on known values. Single-factor redundancy experiment: corrupted adoption 0.900

(sole-sourced) against 0.450 (second seat assigned); clean adoption 0.000 against 0.500.

Transport. 144 runs, 18 cases, two writers (gpt-oss:20b, phi4:14b), two corpora (original, 9 cases; extension, 9 new cases). Supply is judge-measured with inter-judge $r = 0.528$; correcting for attenuation on the original corpus moves 0.150 to 0.217.

Seating gate. Ten candidates; two screen items disjoint from the pipeline case set; 800 visible-character degeneracy threshold; verdicts committed before any pipeline run.

Planner. Identification against analyst-authored labels on the case battery; assignment over 40 plans; live conveyance across briefed seats; end-to-end clean adoption on the pipeline cases.

Out-of-band channel. Carriage: byte-identical prose in both arms, appendix assembled from both-judge caveat sentences; preference 65/116 ties. Uptake: 18 authored items, 11 retained after a registered floor gate, 440 reads, direction-balanced (blocking and enabling caveats), two-token reader decisions.

Re-consultation. Six planted tensions; conditioned n of seven in the scripted decisive cell; judge agreement 0.703 on the analysis population and 0.583 overall, both reported. Live cells compare against archived arms under a registered, never-pooled fresh-batch check. Fabrication follow-up: 20 replies with the fact in hand, 100 elsewhere.

Head-to-head. 108 pairs; 54 live harness pipelines; two judge families; order-debiased.

C. Instrument Validation

Fourteen instrument-validity findings are recorded in the ledger. The three that bear on this paper: (i) entanglement, exposed by a pattern-level decomposition in which an instruction moved dictated phrasings from 0/30 to 28/30 and non-dictated phrasings of the same construct from 1/30 to 4/30 (difference-in-differences +0.83 [+0.67, +1.00]); (ii) the dictation registry, 125 entries from 35 prompt symbols by AST parsing and quote scanning, which on its first run found an earlier judge prompt enumerating the seven phrases the writing prompts dictated, and whose measured over-attribution is zero across 2,907 spans (438,797 characters) and 60 de-scaffolded runs; and (iii) reading-order dominance, with raw first-position preference of 0.85 to 0.88 across a 20B MoE, a 30B MoE, a 14B dense model, and a 7B dense model, and decisive rates of 0.05 to 0.40 after debiasing. Document-level construct judging scored 0.622 agreement against 0.910 for the sentence-level protocol and was rejected.